UMAIR HUSSAIN, PhD

Data Scientist | Physics-based Modeling | AI/ML for Energy

husain.umair2010@gmail.com ☎ +91 9560842249

umairhussaincmm.github.io **LinkedIn:** linkedin.com/in/umair-h-204191b4 **GitHub:** github.com/umairhussaincmm

PROFESSIONAL SUMMARY

Data Scientist at ChampionX, an SLB company, focused on developing AI, machine learning, and physics-based solutions for the oil & gas industry. PhD in Mechanical Engineering from IIT Madras with deep expertise in the finite element method (FEM), phase field modeling, and multiphysics simulation. Currently leading multiple AI-on-Edge initiatives for ESP diagnostics and autonomous optimization, and specializing in integrating data-driven AI with physics-informed and FEM-based modeling to solve complex engineering challenges across artificial lift systems (sucker rod pumps, ESPs), corrosion prediction, and equipment diagnostics. Proficient in Python, C++, deal.II, TensorFlow, and modern ML frameworks, with hands-on experience across the full modeling lifecycle from data curation and SME-driven validation to deployment.

TECHNICAL SKILLS

Machine Learning & AI: Supervised and Unsupervised Learning, Time-Series Forecasting, Time Series Foundation Models, Anomaly Detection, Deep Learning, Gaussian Process Regression, Small Language Models (SLMs), LLM fine-tuning, Physics-Informed ML

Physics-based Modeling: Finite Element Method (FEM), Phase Field Modeling, Multiphysics Simulation, Computational Mechanics, Plasticity, Electrochemical Modeling, Multi-phase Field Solvers

Programming: Python (NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow), C++, MATLAB, LaTeX, Shell/Bash

Tools & Libraries: deal.II, TensorFlow, PyTorch, Abaqus, MicroSim, ParaView, GMSH, SageMath, PETSc, Git, Linux

AI Tools: Cursor, GitHub Copilot, ChatGPT, Claude

Domain Expertise: Artificial Lift Systems (Sucker Rod Pumps, ESPs), Corrosion Prediction, Equipment Diagnostics, Sensor Data Analytics, Li-ion Battery Modeling

Soft Skills: Cross-functional collaboration, Technical writing, Research mentoring, Presenting to technical and business audiences

PROFESSIONAL EXPERIENCE

Data Scientist*, ChampionX, an SLB company* May 2025 – Present

Chennai, India

- Developed **AI, machine learning, and physics-based models** across multiple oil & gas product lines including **artificial lift systems** (sucker rod pumps, ESPs), **corrosion prediction**, and **equipment diagnostics**.
- Led two flagship **AI-on-Edge initiatives**, the **Autonomous ESP Optimizer** (patent application submitted) and a **Smart Alarm model upgrade** (currently under deployment), and supported the **SLM Agent for ESP diagnostics** initiative.
- Integrated **physics-informed modeling** (FEM, viscoelasticity, multiphysics) with modern ML/DL and generative AI (SLMs) to deliver interpretable, high-fidelity models suitable for field deployment.
- Owned the end-to-end modeling workflow (problem framing, data curation and labelling, model development, SME-driven validation, and hand-off for productionization), and produced a labelled ESP dataset now reused across the team.

Research Analyst, Intern*, ChampionX, an SLB company* Nov 2024 – Apr 2025

Chennai, India

- Contributed a **finite element (FEM) model of the sucker rod pump for deviated wells** to the team's ongoing pump modelling effort, a novel extension over standard vertical-well models, to predict downhole pump cards and support diagnostics under realistic well trajectories.
- Developed an **ML-based broken-shaft prediction model** from operational sensor data to support predictive maintenance and reduce unplanned equipment failures.

SELECTED PROJECTS AT CHAMPIONX, AN SLB COMPANY

Autonomous Optimizer for ESP *Autonomous Control · Optimization · Production Maximization*

Developed a data-driven optimizer that automatically identifies the setpoint required to maximize production from an Electric Submersible Pump (ESP). The system is currently under testing, and a patent application for the approach has been submitted to the company's legal team.

Smart Alarm Model Upgrade for ESP Diagnostics *Anomaly Detection · ESP Diagnostics · Production Deployment*

Designed a new substitute model to replace one of the existing smart alarms used for ESP diagnostics, specifically the model responsible for detecting ESP or well stabilization after a restart. The new model is currently under deployment, and work is ongoing to identify further improvements to other smart alarms in the suite.

SLM Agent for ESP Diagnostics *Generative AI · Small Language Models · Knowledge Engineering*

Contributed to an SLM-based agent for ESP diagnostics by building the underlying knowledge bank and curating a suite of test cases drawn from the internal labelled ESP dataset and other sources. Worked closely with an SME to analyze and interpret model results during the ongoing testing phase.

Gas-Interference Anomaly Detection in ESPs *Anomaly Detection · High-Fidelity Sensor Data · Data Labelling*

Developed an anomaly detection model targeted at gas interference in ESPs using high-fidelity operational data. Partnered with a subject-matter expert to test and validate the model over one month of data across 16 ESPs, in the process creating a labelled ESP dataset that has since been reused by the team for multiple downstream modeling and testing efforts.

FEM Model for Sucker Rod Pumps in Deviated Wells *FEM · Physics-based Modeling · Artificial Lift*

Supported an ongoing team effort on sucker rod pump FEM modelling by contributing the deviated-well version of the model, and subsequently by incorporating viscoelastic material behaviour to capture time-dependent rod-string response and improve fidelity of pump card prediction under realistic downhole conditions.

CO_2-based Corrosion Prediction: Hybrid vs XGBoost *Hybrid Physics + ML · Benchmarking · Materials & Reliability*

Benchmarked a hybrid physics-informed machine learning model against an XGBoost baseline for CO_2-based corrosion prediction in oil & gas assets.

DOCTORAL RESEARCH EXPERIENCE

Ph.D. Thesis: Electrochemical Response during Lithiation of High-Capacity Anodes in Li-ion Batteries

Supervisors: Dr. Narasimhan Swaminathan & Dr. Gandham Phanikumar, IIT Madras

- Developed a **multiphysics phase field model** coupling chemical diffusion under stress with electrochemical reactions via Butler-Volmer boundary conditions to study anode (dis)charging behavior.
- Demonstrated the **impact of phase transformation, electrode size, and stress state** on electrochemical response through voltammogram analysis, including both elastic and elasto-plastic regimes.
- Built a **multi-phase field solver** for systems with multiple phases; established a physically grounded mapping between model parameters and real material properties.
- Applied the multi-phase field framework to **peritectic solidification**, showing how parameters should be selected for a given material system.
- Implemented all solvers using the C++ FEM library **deal.II**; contributed a standalone solver to the deal.II code gallery.
- Achieved **~60% performance improvement** through parallelization with PETSc-based wrappers in deal.II.

Data-driven Modeling of Li-ion Battery Electrode Microstructures

Collaborators: Dr. Anand Agrawal & Dr. Narasimhan Swaminathan

- Predicted **electrode performance** from microstructural features (e.g., grain size) to support battery electrode design optimization.
- Generated voltammogram data with a deal.II FEM solver, built microstructure meshes with Microgen and GMSH, and applied **Gaussian Process Regression** to learn a nonlinear mapping between voltammogram features and grain size.

EDUCATION

M.S. + Ph.D., Indian Institute of Technology Madras 2019 – 2025

Mechanical Engineering | CGPA: 9.39

B.Tech, Jamia Millia Islamia 2015 – 2019

Mechanical Engineering | CGPA: 9.66

PUBLICATIONS

Journal Articles

Hussain, U., Phanikumar, G., & Swaminathan, N. (2023). Mapping of multiphase field model parameters to physical factors in order to simulate desired phase transformations. *Computational Materials Science, 226, 112227. https://doi.org/10.1016/j.commatsci.2023.112227*

Hussain, U., Swaminathan, N., & Phanikumar, G. (2024). Numerical Voltammetry of Phase Separating Materials Using Phase Field Modeling. *Journal of The Electrochemical Society, 171(7), 070502. https://doi.org/10.1149/1945-7111/ad59cc*

Agrawal, A. K., Swaminathan, N., & Hussain, U. (2025). Probabilistic Modelling of the Nonlinear Mapping between Voltammograms and the Grain Size of the Electrode Material in Li-Ion Batteries Using Optimally Tuned Gaussian Process Regression Models. *Journal of The Electrochemical Society, 172(1), 013501. https://doi.org/10.1149/1945-7111/ada0b6*

Conference Proceedings

Hussain, U., Swaminathan, N., & Phanikumar, G. (2024). A Multi-phase Field Framework Application to the Study of the Peritectic Reaction for an Arbitrary Material System. In *Recent Advances in Mechanics of Functional Materials and Structures (pp. 231–240). Springer Nature Singapore. https://doi.org/10.1007/978-981-99-5919-8_20*

Open-source Contributions

Hussain, U. Crystal growth using phase field modeling. In *deal.II Code Gallery. https://dealii.org/developer/doxygen/deal.II/code_gallery_Crystal_Growth_Phase_Field_Model.html*

CONFERENCE PRESENTATIONS

16th World Congress on Computational Mechanics (WCCM 2024) 21 – 26 July 2024

Vancouver, Canada

XVII International Conference on Computational Plasticity (COMPLAS) 5 – 7 Sept 2023

Universitat Politècnica de Catalunya, Barcelona, Spain

8th Asian Conference on Mechanics of Functional Materials & Structures 11 – 14 Dec 2022

IIT Guwahati, India

ME@75: Research Frontiers 29 June – 1 July 2022

IISc Bangalore, India

TEACHING EXPERIENCE

Doubt-session Instructor, NPTEL Jul 2022 – Nov 2023

Basics of Finite Element Analysis I & II; FEM: Variational Methods to Programming.

Short Course Instructor, PSG Institute of Technology Feb 2022 – May 2022

Short course on Computational Tools.

ACADEMIC PROJECTS

- **Applied FEM:** Solved the SIR model for disease spread in Uttar Pradesh using a self-developed FEM code in Python with Picard iteration, an implicit time-marching scheme, and GMSH meshes, achieving steady-state at 87 days.
- **Engineering Plasticity:** Implemented the non-associative Drucker–Prager plasticity model for pressure-sensitive materials in a self-developed FEM code in MATLAB, derived the tangent modulus, and validated the model against the J_2 formulation.

AWARDS, FELLOWSHIPS & WORKSHOPS

Awards & Achievements

- 2020: Awarded the prestigious **Prime Minister Research Fellowship (PMRF)**.
- 2017: B.Tech scholarship awardee (second year).
- Contributed a standalone solver to the **deal.II open-source FEM code gallery**.

Fellowships & Grants

- 2020 – 2024: **Prime Minister Research Fellowship (PMRF)**. Fellowship: INR 36,60,000; Research Contingency: INR 8,00,000.
- 2019 – 2020: **Half-time Research Assistantship (HTRA)**. INR 3,72,000.

Workshops

- 2023: Technical and Scientific Writing, IIT Madras.
- 2020: Faculty Development Program on Advances in Theoretical and Computational Mechanics, NIT Andhra Pradesh.
- 2017: Basics of Design and Innovation, Centre for Innovation and Entrepreneurship, Jamia Millia Islamia.